```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Dataset Path (use your own path from Google Drive)
dataset_path = "/content/EmotionDataset"

# Image preprocessing
datagen = ImageDataGenerator(
    rescale=1/255.0,
    validation_split=0.2
)

train_data = datagen.flow_from_directory(
    dataset_path,
    target_size=(48, 48),
    color_mode='grayscale',
    batch_size=32,
    class_mode='categorical',
    subset='training'
)

val_data = datagen.flow_from_directory(
    dataset_path,
    target_size=(48, 48),
    color_mode='grayscale',
```

```
    batch_size=32,
    class_mode='categorical',
    subset='validation'
)

# ---------------- CNN MODEL FROM SCRATCH -----------------
model = tf.keras.models.Sequential([

    # 1st Convolution Block
    tf.keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(48,48,1)),
    tf.keras.layers.MaxPooling2D(2,2),

    # 2nd Convolution Block
    tf.keras.layers.Conv2D(64, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),

    # 3rd Convolution Block
    tf.keras.layers.Conv2D(128, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),

    # Flatten + Dense layers
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dropout(0.5),
```

```
    # Output layer = number of classes
    tf.keras.layers.Dense(train_data.num_classes, activation='softmax')
])

# Compile the CNN
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train the model
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=20
)

# Save model
model.save("emotion_cnn_model.h5")
import cv2
import numpy as np

img = cv2.imread("/content/test.jpg", cv2.IMREAD_GRAYSCALE)
img = cv2.resize(img, (48, 48))
```

```
img = img.reshape(1, 48, 48, 1)/255.0

prediction = model.predict(img)
emotion_index = np.argmax(prediction)

classes = list(train_data.class_indices.keys())

print("Detected Emotion:", classes[emotion_index])
```